

Proposal

Reducing Curtailment through Smart Appliances & Boilers

Leyla Mabey - 02281847

Krzysztof Wancerski - 02201834

Arancha Ramirez - 02122361

Emily Haynes - 02215562

Josh Williams - 02086121

Zoe Hall - 02269583



Background context

Reducing Curtailment through Smart Appliances & Boilers

WHERE?



The Orkney Isles

WHO?



Kaluza,
Government,
Energy Suppliers,
Households

WHAT?



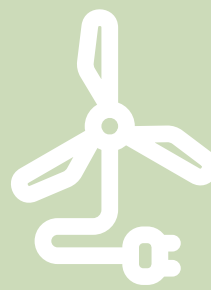
Demand
Response
Program

WHEN?



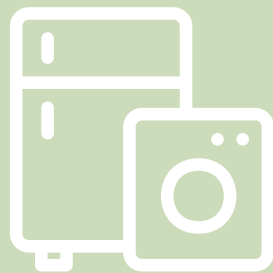
2026

WHY?



Bridge the gap
between wasted
energy and
households.

HOW?



Smart
Appliances

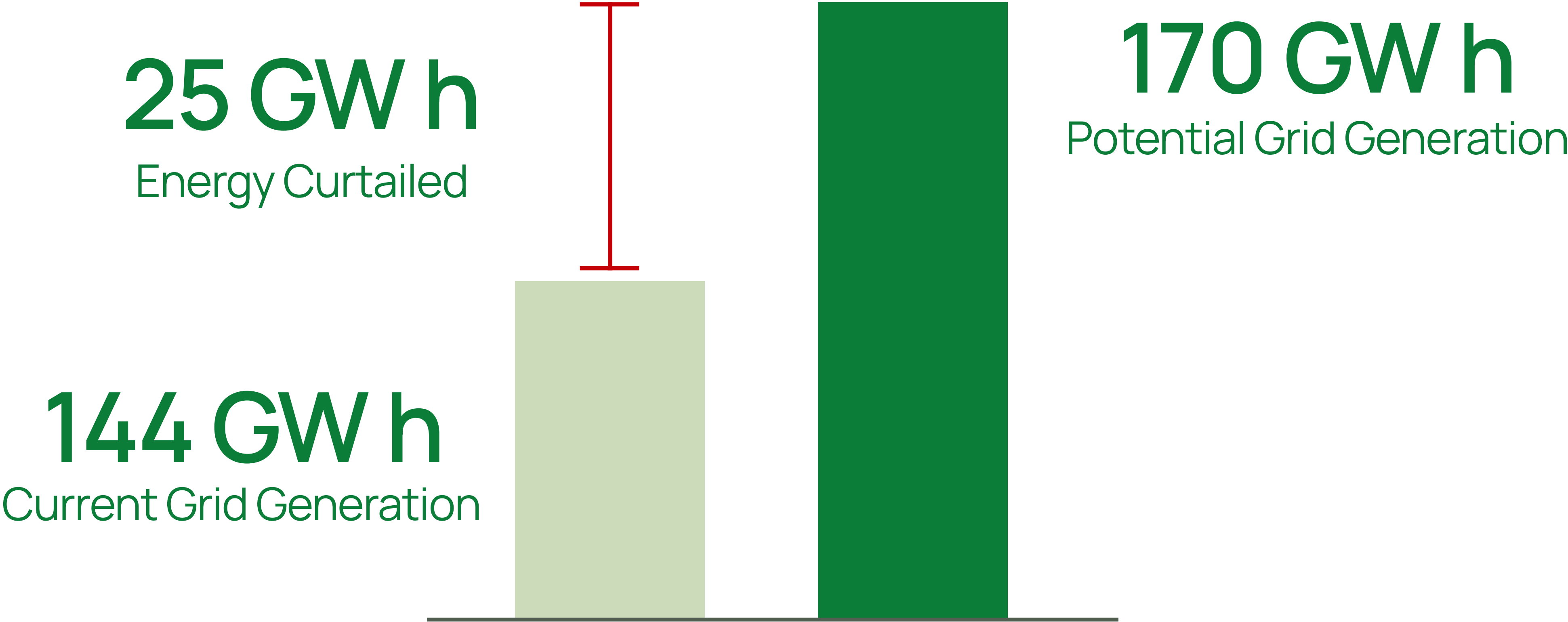
The Orkney Islands

With over twenty-two thousand inhabitants, the Orkney Islands sit in an unusual energy position.



22K inhabitants
63% in fuel poverty
23 turbines

Surplus Energy

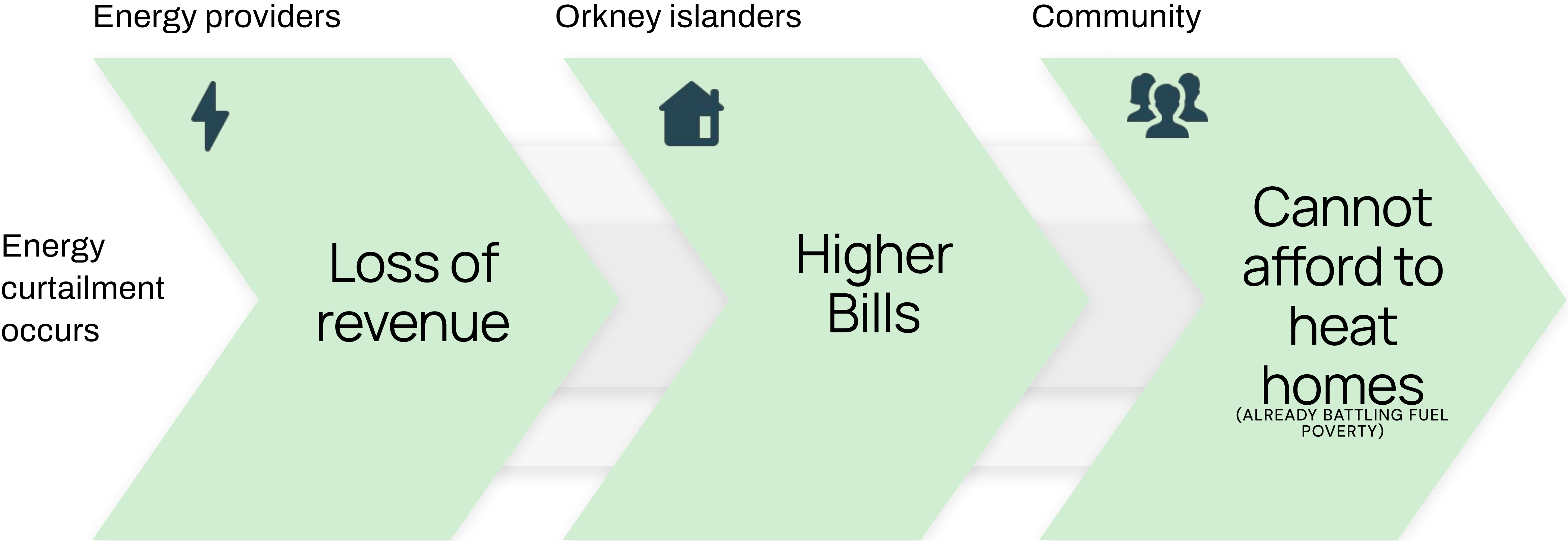


Annually Missed Revenue

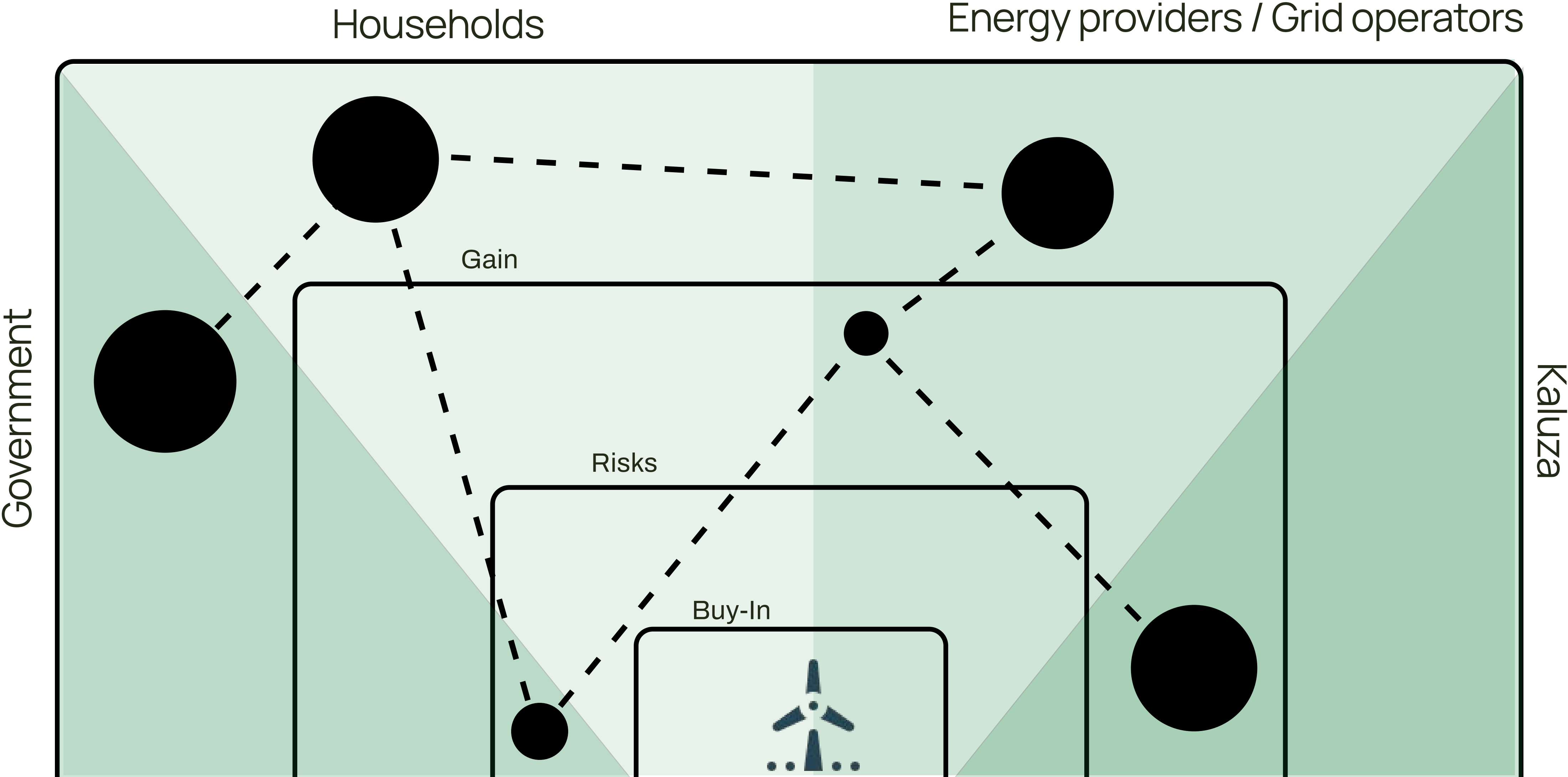
- Discarded Energy Payments
- Annual Energy Wasted



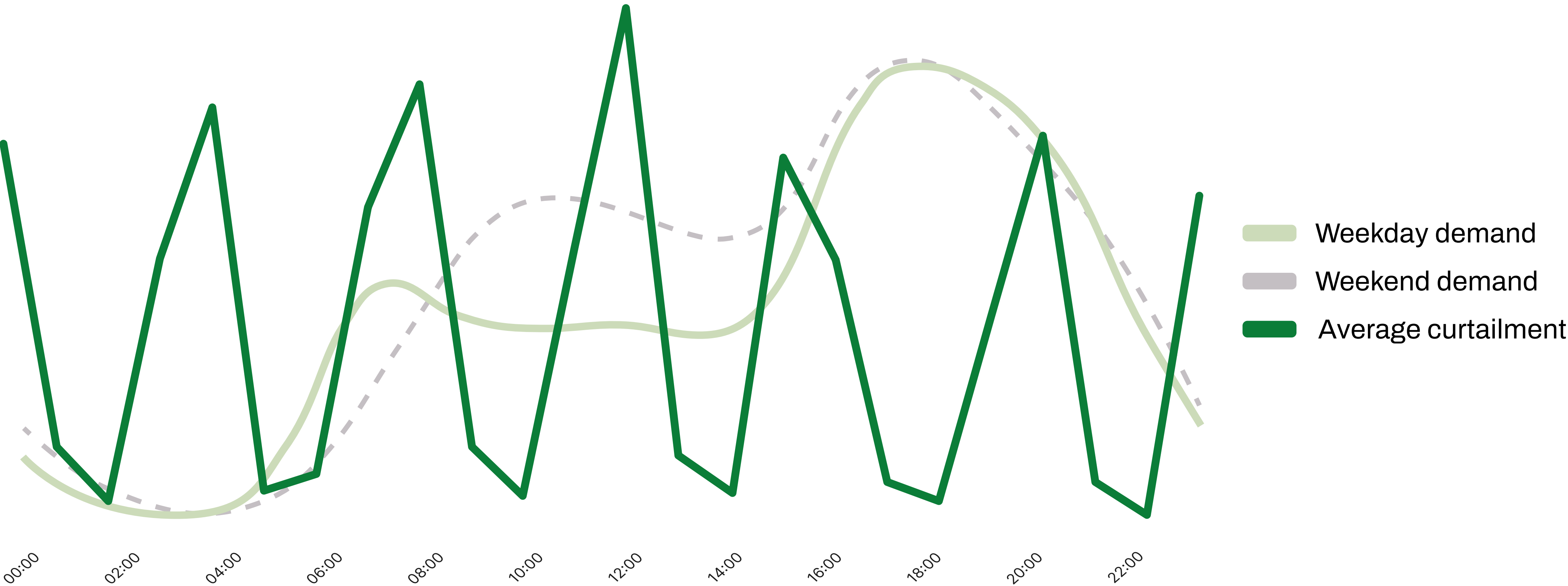
Negative consequences for the island



Stakeholders hold different priorities



... And Different Behaviours

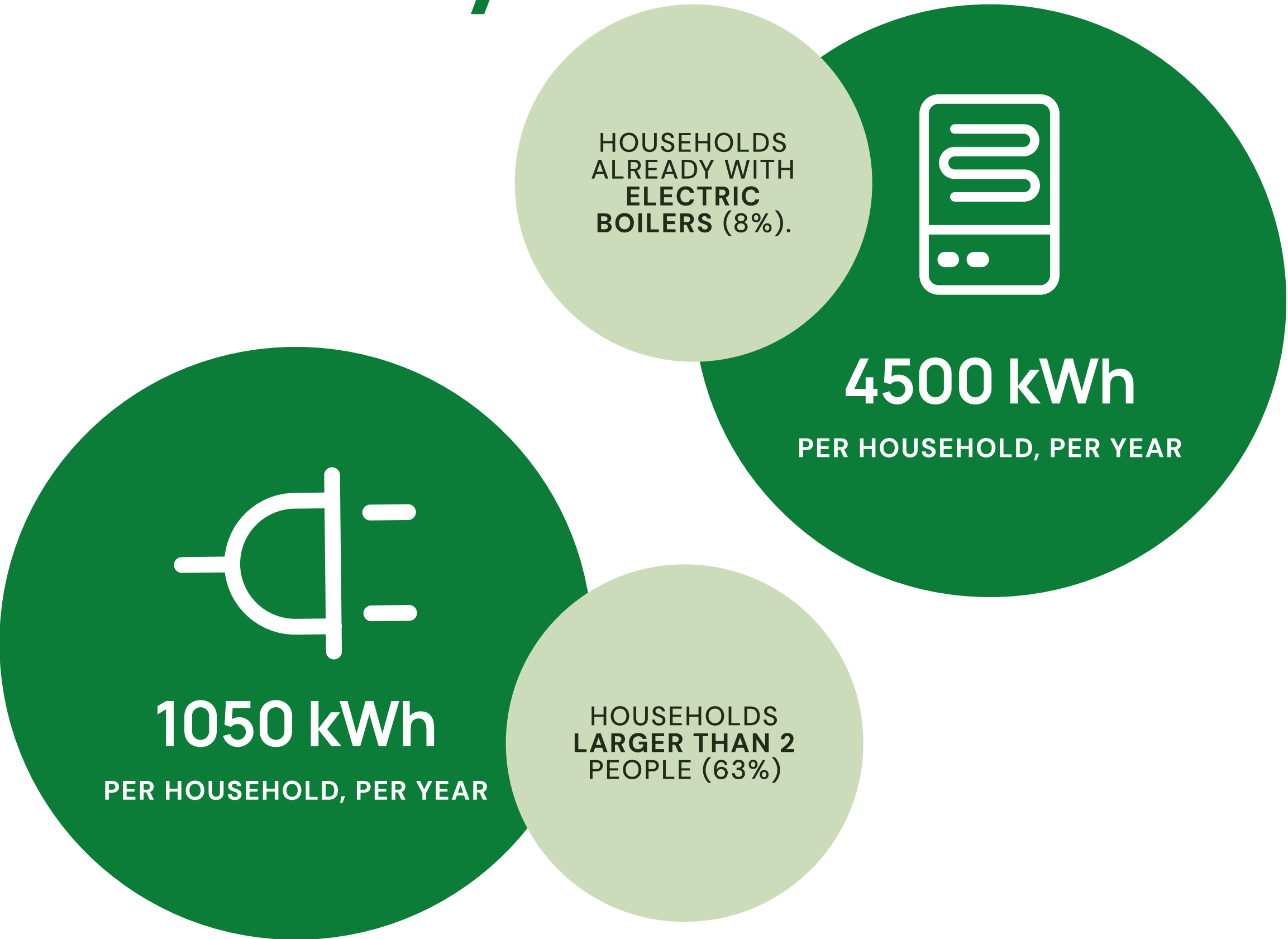


Rigid evening demand
(cooking, entertainment)

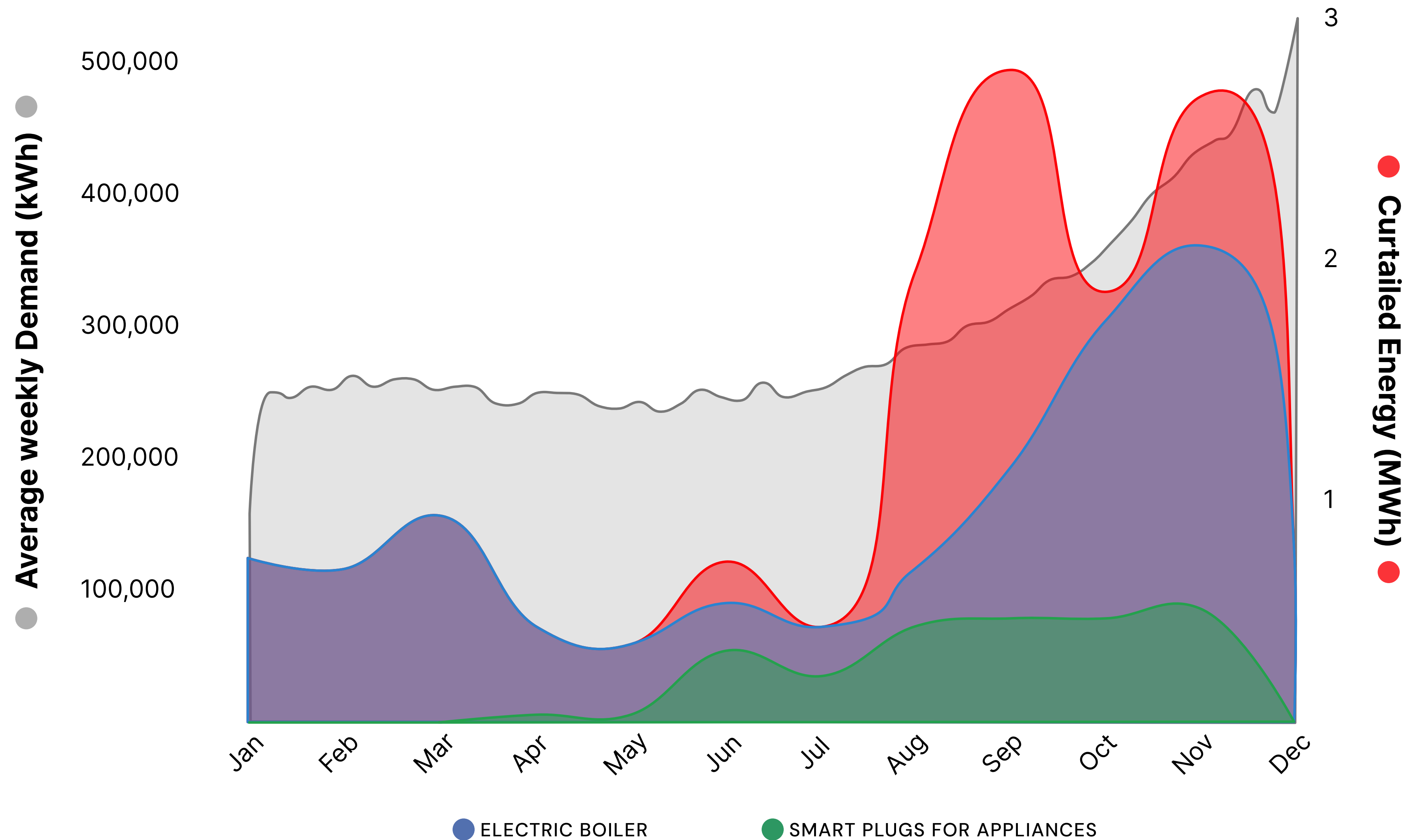
Unused flexible **capacity**
overnight

The **workday gap** is
untapped storage

5550 kWh per household shifted annually



Solving seasonality



Demand Response Results!



£2.5 MILLION LOST FROM CURTAILMENT

72.2% of curtailed energy mitigated

PILOT LAUNCH!

£80-150 RANGE LIMIT

INSTALLATION COST PER HOUSEHOLD

£ 90

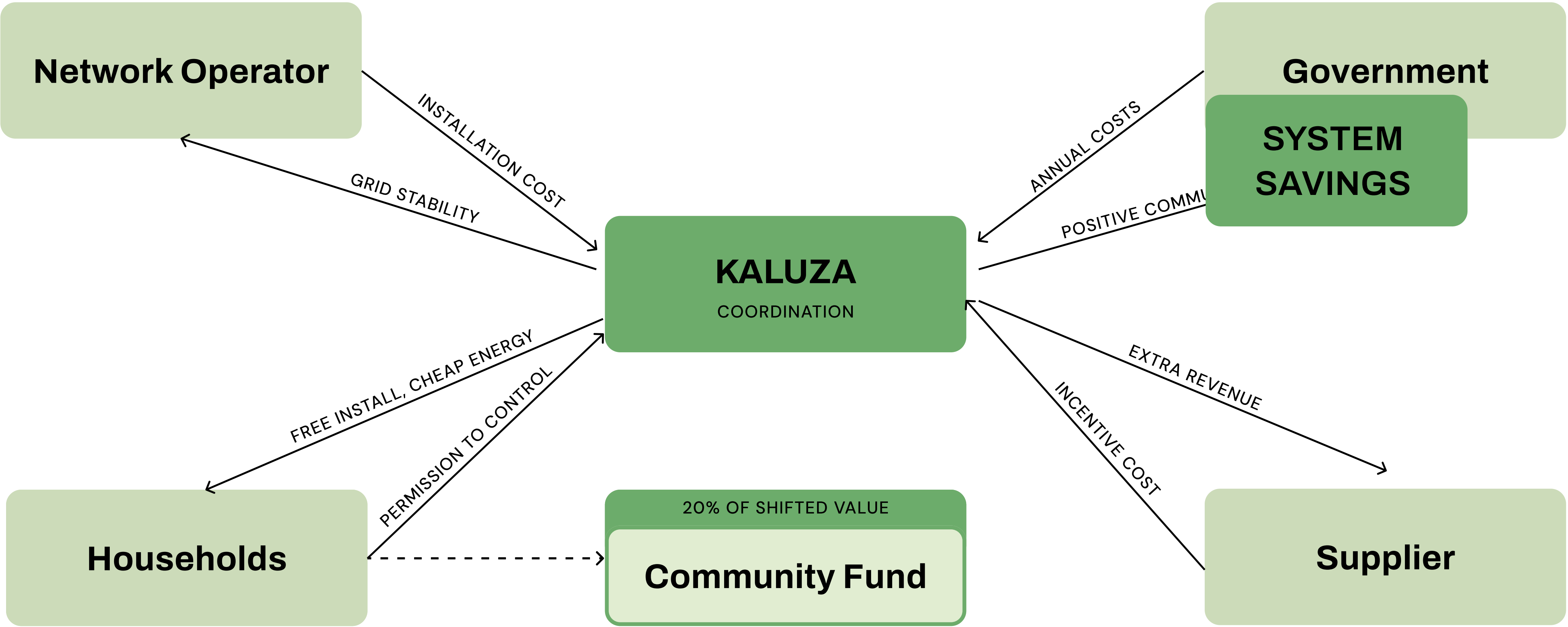
PAID BY THE OPERATOR

40% MINIMUM PARTICIPATION

71%

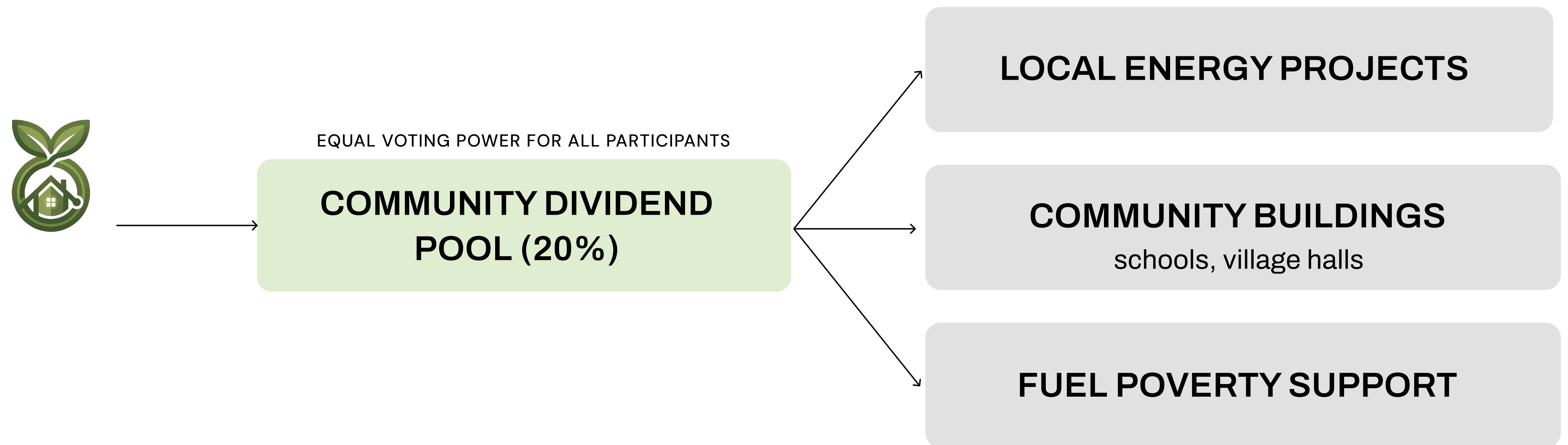
ACCESSIBLE HOUSEHOLDS

Simple & Fair Pricing Strategy



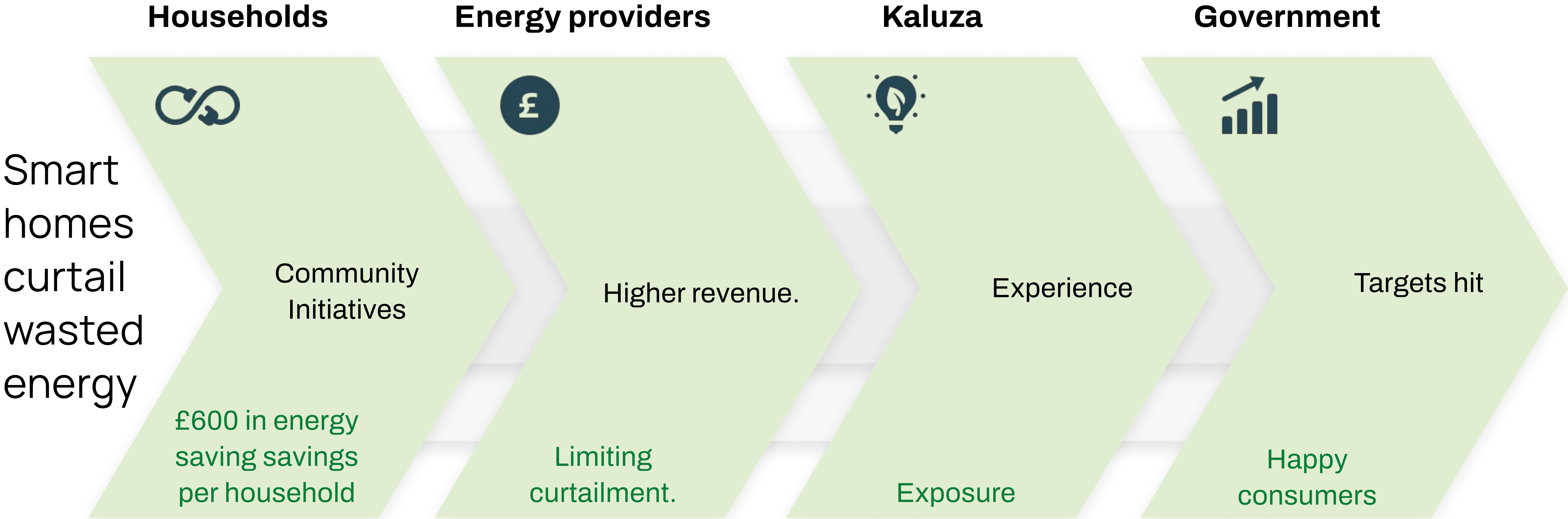
Ensuring Fairness: Community Fund

→ The Mechanism



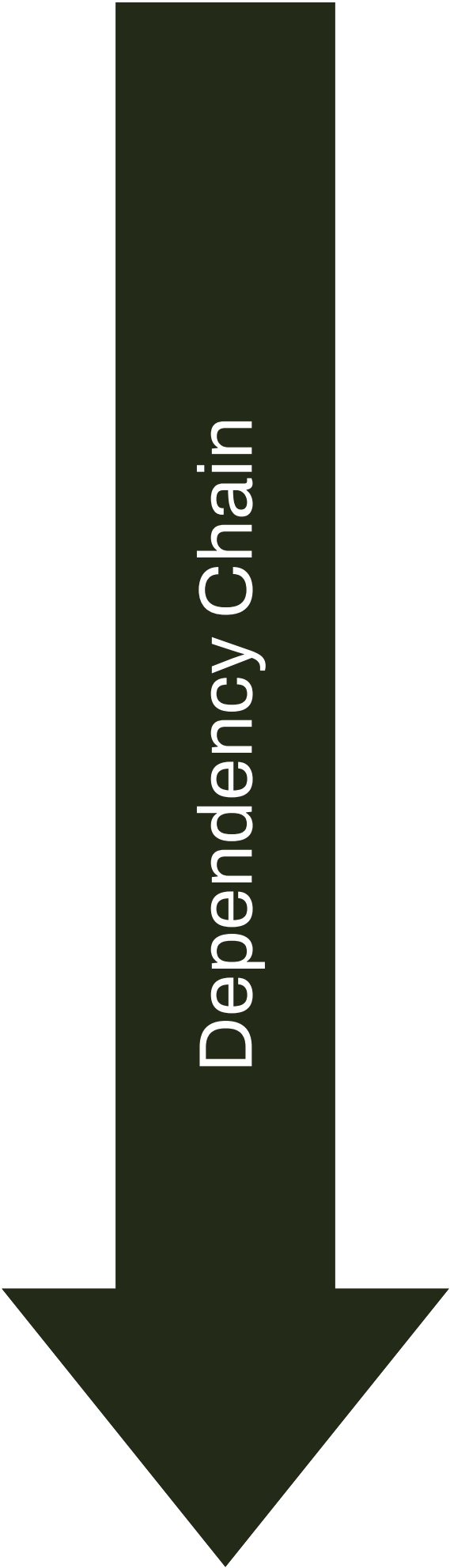
Positives for every stakeholder

→ What do stakeholders gain from our solution?



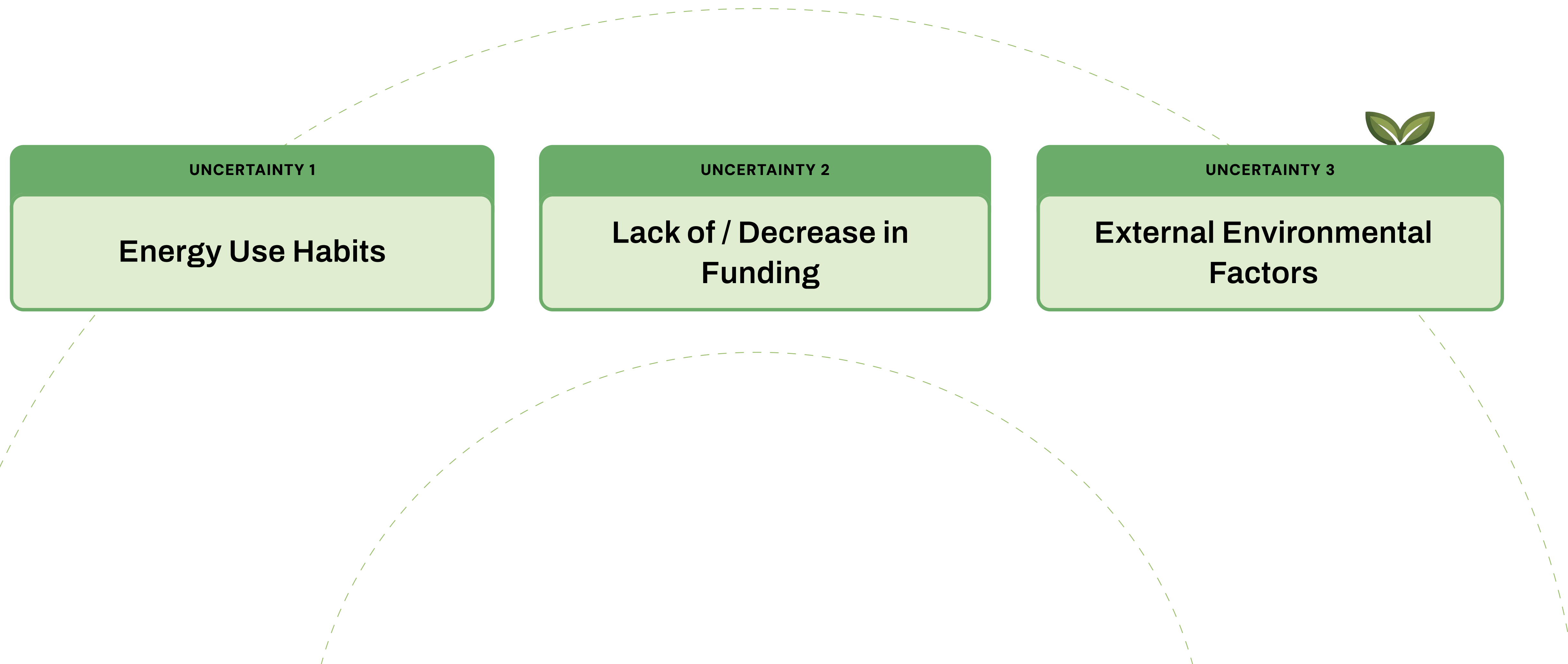
Risks are dependent

	LIKLEHOOD	IMPACT	MITIGATION
Regulatory stakeholders block the programme REGULATION	Low	High	Involve a consumer protection stakeholder in the incentive design process
Participation Rates are too low SOCIAL	Med	Med	Define minimum viable household threshold before recruitment opens, with backup recruitment pool.
System doesn't perform during peak activation TECHNICAL	Med	High	Build redundant communications pathways, so that a single point of failure cannot destroy performance

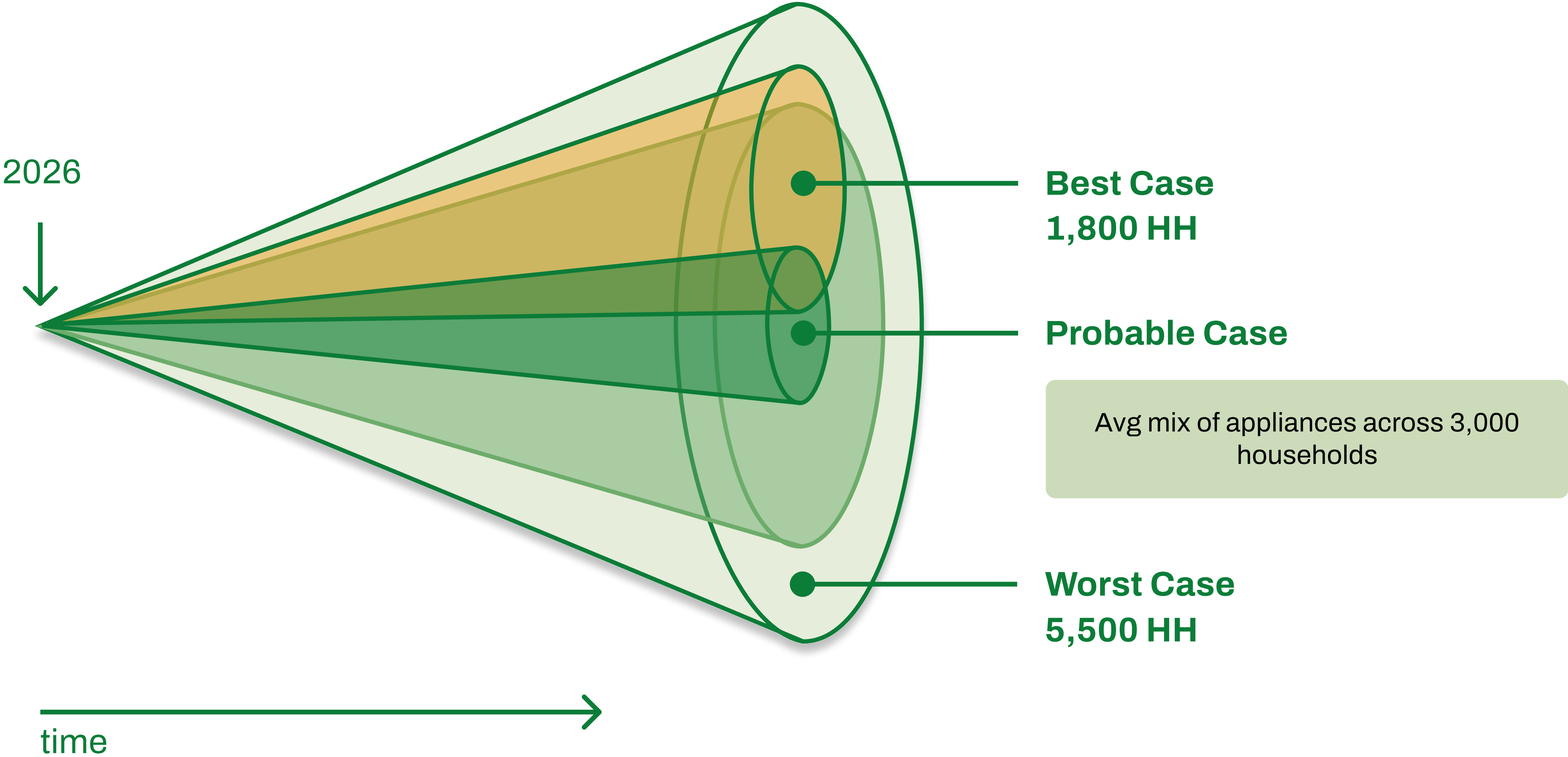


Uncertainty Analysis

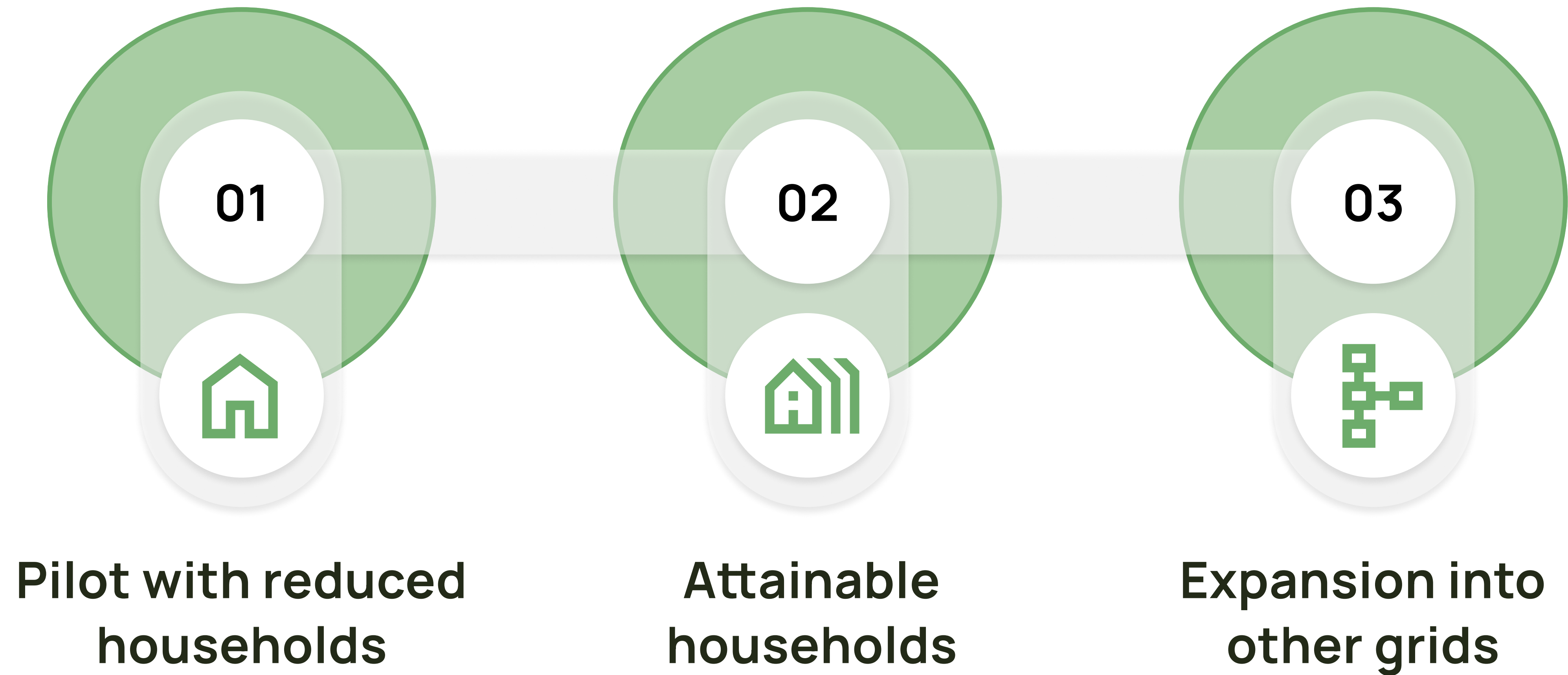
→ What are the largest uncertainties in our system?



Ensuring the £2.5M Recovery

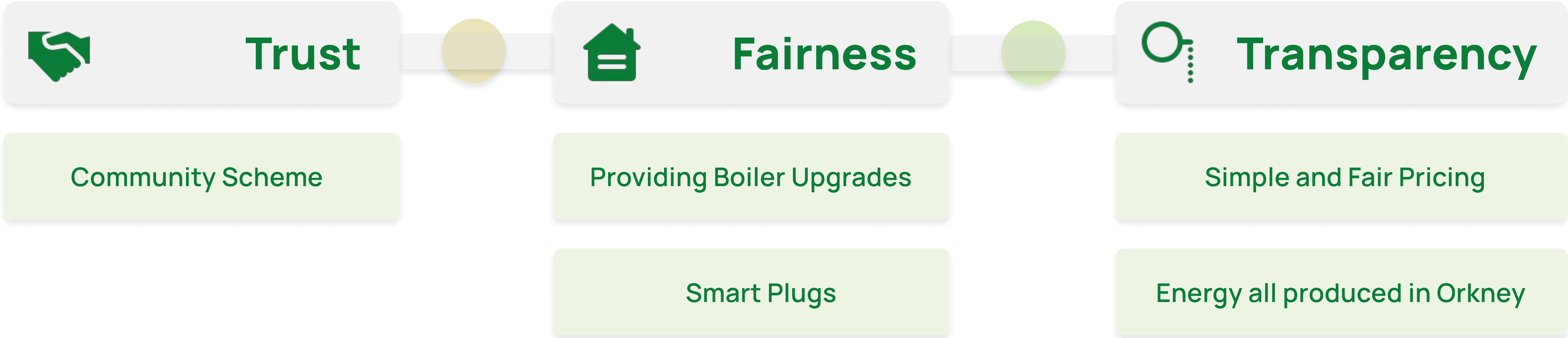


From Pilot to Scalable Market



Values

→ Developing a resilient and fair system.



Why this proposal?

Reducing Curtailment through Smart Appliances & Boilers



Meets the needs of all stakeholders.



Risk Aware, Flexible Strategy balancing tradeoffs.



Successfully reduces curtailment by 72.2% (£1.8M)



Paves the way for future implementation.



Reduced household uptake can decrease the impact



APPENDIX



OLIVE

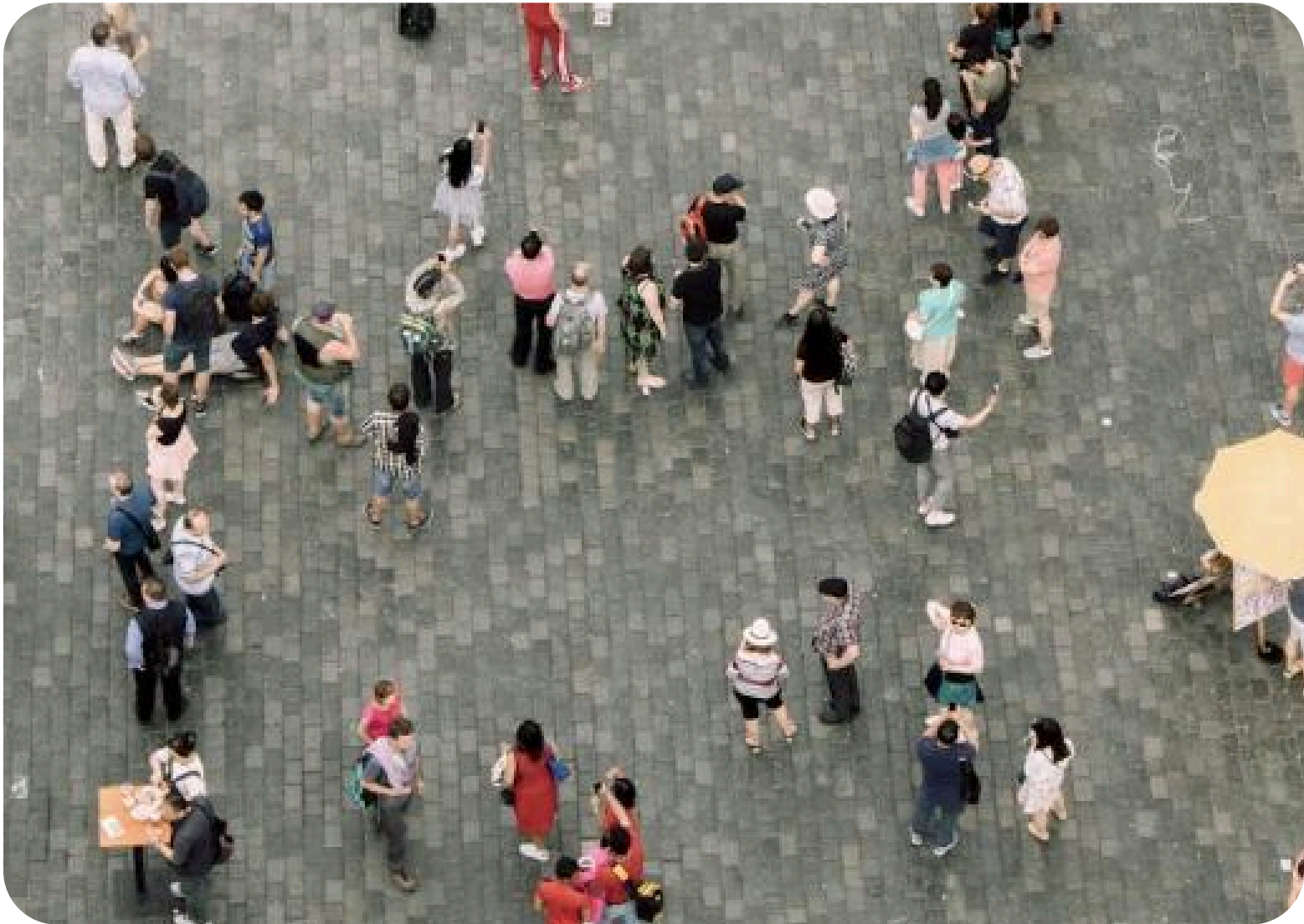
Powering Smarter Energy Decisions

DR Program Benefits



Ecofriendly Energy Efficiency

This program drastically reduces green energy waste. By implementing the DR program and making wind energy cheaper, we position green wind energy as the more attractive option to consumers.



Community Focused

Olive Consultants prioritises equitable distributions of resources. The program ensures part of the savings go directly into initiatives supporting all of the community such as car parks or playgrounds.



Stakeholder Alignment

The program provides clear value across the board: households see lower bills, suppliers gain grid control, councils strengthen community resilience, and the Government accelerates its Net Zero targets.

A deeper dive into the energy paradox



Fuel Poverty

As of 2025, roughly 31% of households are classed as being in fuel poverty. These households spend more than 10% of their income on heating. This is higher than the Scottish average of 24%. The number of households spending over £2,500 annually on electricity has quadrupled since 2019. (European Marine Energy Centre, 2024)



Excess wind energy

While residents struggle with bills, the islands' wind turbines are frequently forced to shut down. Community turbines have on certain years faced a 30% loss in production. Orkney generates 128% of its own needs, yet this excess energy is often unrealised because there is no local demand-side flexibility. (Islands Centre for Net Zero)

Stakeholder Advantages

→ What do stakeholders gain from our solution?

Consumers/Community

- Energy cost saving ~ **5600kwh** shifted per house
- Community Benefits Schemes

Energy providers

- Prevents local substation overloads by using households as an active sink during peak wind generation.

Kaluza

- Experience in implementing a demand response system

Government

- In line with strategic objectives of reducing wasted energy and developing renewable energy solutions

Calculating Curtailment

Curtailment: Periods where turbine's Setpoint_kw was restricted below its max capacity of 900kW.

Assumptions

- Filtering: We only counted intervals where actual power generation was within 1% of the restricted point.
- Lost Energy = (900kW - Setpoint) x Duration.
- Fleet size: 23 turbines.
- Financial Value of energy = £0.10 kWh.
- Assume single turbine is representative of fleet's behaviour.

Single Turbine

Annual Energy Waste = 1,114,354 kWh (1.1GWh)

Annual Value Lost = 1,114,354 kWh x £0.10 = £111,435

Fleet Opportunity

Total Unrealised Power = 1.1GWh x 23 = 25.6GWh

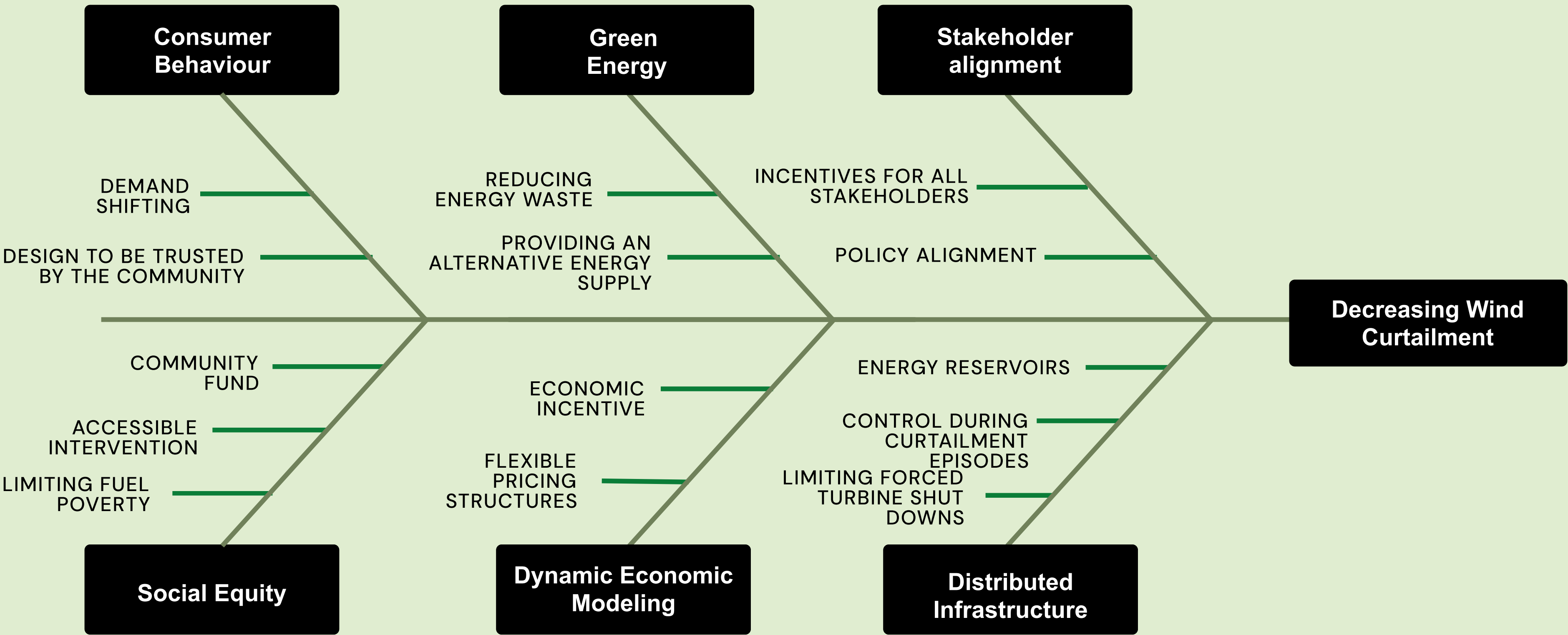
Total Financial Opportunity = 25.6GWh x £0.10 = £2,563,014



TOTAL LOST VALUE OF CURTAILMENT:
£2.5 Million annually.

Eliminating Curtailment

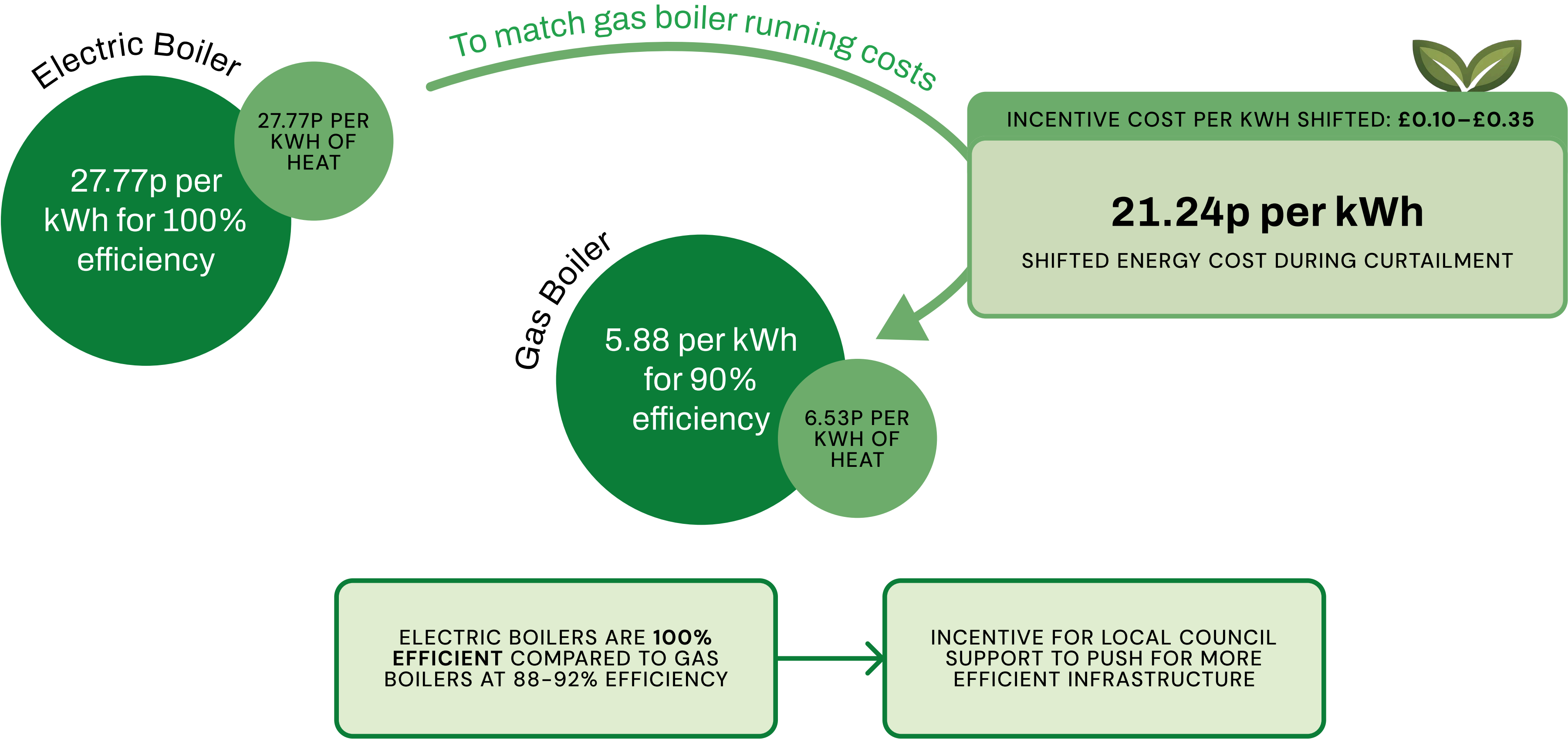
Ishikawa Diagram: Multilateral Solutions for Demand Response Program.



Electric boiler proposition

Offering electric boiler discounted rates guaranteeing electricity costs that match or are lower than gas costs, to encourage conversion to this scheme. This is given that the household runs the boiler flexibly, only at times of energy curtailment.

Prioritising discounted energy rates to electric boilers over smart home appliances.



Electric boiler proposition

Electric boilers are cheaper to purchase than gas, but due to the cost of electricity, are more expensive to run long term. they are often suitable for smaller properties (1 bed, 1 bath).

Gas boilers		Electric boilers	
10-15 years	Lifetime	15-25 years	REQUIRES A GUARANTEE FOR HOW LONG REDUCED COSTS DURING CURTAILMENT WILL LAST
£620-£1260	Current running costs	~£1,700	
Risks of CO leaks	Safety	No risk of CO leaks	
Requires flue installation	Installation	No flue requirement	

PIPE TO SAFELY EXPEL DANGEROUS COMBUSTION GASES

SMART PLUGS: 2,254 HOUSEHOLDS (10% OF POPULATION) – ELECTRIC BOILERS: 676 HOUSEHOLDS (3% OF POPULATION) – TOTAL PILOT: 2,930 HOUSEHOLDS
OVERALL PILOT PENETRATION: 13.0% OF THE POPULATION

Uncertainty Analysis



UNCERTAINTY 1

Energy use habits

- Reliability drops during high-demand windows (e.g., morning routines).
- Users prioritise immediate comfort/hot water, manually bypassing smart controls.
- Frequent interventions can fatigue consumers. Over time, users may ignore notifications or disable smart features entirely to avoid constant habit adjustments.

UNCERTAINTY 2

Lack of/Decrease in Funding

- Current Roadmap assumes that the government will cover costs of upgrading peoples boilers.
- If stakeholder funding decreases or is diverted to different energy schemes, this risks entire business model.
- Electric boilers need to be replaced ~ every 20 years, if funding decreases significantly, we will not be able to proceed with this.

UNCERTAINTY 3

External Environmental Factors

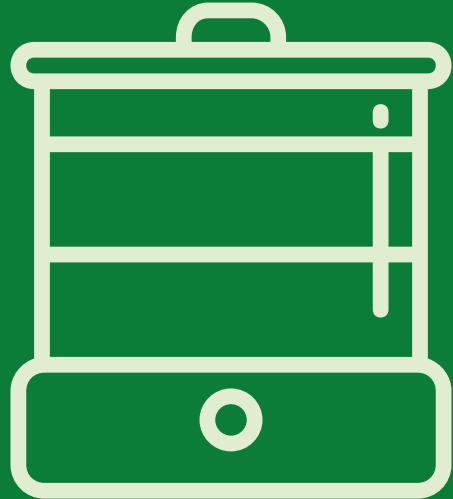
- Curtailment is not a constant. In a low-wind year, the number of DR events could drop significantly.
- Very high wind events could create spikes in waste that overwhelm the sink capacity.
- Seasonal influence may not match prediction. High wind (supply) often occurs during mild, wet autumns. Here, heating demand is lower than in a dry, freezing winter.

Alternative Directions

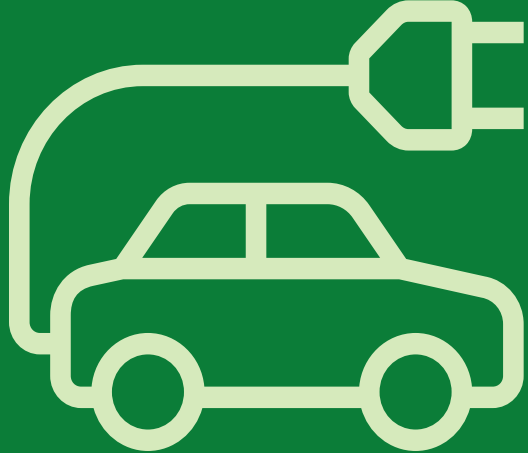
Our team deliberately considered other directions and ways of reducing curtailment. Smart appliances & smart boilers were finally selected as they aligned best with user insights.



Smart Appliances



Smart Boilers



Smart EVs & E-Bikes



Mini Data Centres.

↖ Combined ↗
approach selected

↖ Limited ↗
accessibility

Tradeoffs.

Community Fund vs. Individual Savings

We are diverting 20% of the system value into a community fund.

- This ensures those in fuel poverty who can't host high-flex hardware still benefit from the project. It also improves community trust in the project.
- The Con: It slightly dilutes the direct financial "hook" for the early adopters who are providing the most flexibility. Though the small amount of money that they miss out on is unlikely to be significant enough for them to opt out of the scheme.

Automation vs. Behavioural Change

We are opting for automated intervention (Kaluza controlling assets) rather than asking people to manually change their habits.

- This leads to much higher reliability and "soak" potential because the algorithm/system doesn't get tired or forget to turn on the dishwasher.
- The Con: It reduces the "educational" aspect of the project. Residents might save money without actually learning why or becoming more energy-literate, making them reliant on the software rather than mindful consumption.

Immediate Impact vs. Broad Inclusion

We prioritise high-flexibility assets (Electric Boilers) to hit the 25.3 GWh target quickly.

- A smaller number of homes (1,800–3,000) can solve the entire £2.5M wastage problem.
- The Con: Focusing on boilers naturally favours certain types of housing. While smart plugs help with fairness, the "big wins" come from households that have the space and infrastructure for electric heating, potentially creating a two-tier benefit system.

Pricing Breakdown

THE TEAM CONSIDERED THE MOST CONSERVATIVE ESTIMATES FOR PRICES.

Network Operator

PAYS FOR:

- ONE-OFF INSTALLATION (£150 PER HOME)
- 20% OF INCENTIVE COST (£/KWH SHIFTED)

WHY:

- AVOIDED OR DEFERRED GRID REINFORCEMENT
- PEAK SHAVING DURING VOLATILE EVENING DEMAND
- INCREASED PREDICTABILITY OF LOAD

Government

PAYS FOR:

- ANNUAL FIXED OVERHEADS (£250K) THROUGH THE COMMUNITY ENERGY GENERATION GROWTH FUND.

WHY:

- OUR PROPOSAL MEETS THE KEY CRITERIA OF COMMUNITY & SOCIAL EQUITY IN ENERGY SUPPORT.
- AVAILABLE TO APPLY TO THE COMMUNITY ENERGY GENERATION GROWTH FUND.

Generators & Suppliers

CURTAILMENT MONETISATION

PAYS FOR:

- PLATFORM AND SUPPORT COSTS (£40 PER HOME PER YEAR)
- 80% OF INCENTIVE COST (£/KWH SHIFTED)

WHY:

- NEW REVENUE FROM OTHERWISE WASTED GENERATION
- A FLEXIBLE CUSTOMER BASE WITHOUT TRADITIONAL ACQUISITION COSTS

Local Council

EQUITY & CONTINUITY VALUE

PAYS FOR:

- FAIL SAFE: ANNUAL FIXED OVERHEADS (£250K):
- COMMUNITY ENGAGEMENT
 - DATA GOVERNANCE
 - LONG-TERM OPERATIONAL CONTINUITY

WHY:

- GUARANTEES FAIRNESS AND INCLUSION
- ENSURES THE SCHEME RUNS EVEN AT LOW INITIAL UPTAKE
- DELIVERS VISIBLE NET ZERO AND SOCIAL OUTCOMES

Community Fund

DETAILS:

- 20% OF TOTAL SYSTEM SAVINGS DIVERTED FROM INDIVIDUAL REWARDS
- PAID TO ORKNEY COMMUNITY TRUST
- FAIR REDISTRIBUTION.

OUTCOME

- LOW-FLEX HOUSEHOLDS STILL BENEFIT
- COMMUNITY-LEVEL IMPROVEMENTS (E.G. PUBLIC BUILDINGS, FUEL POVERTY MITIGATION)
- ADDRESSES DISTRIBUTIONAL FAIRNESS CONCERNS EXPLICITLY

Consumer Offer

BENEFITS:

- CHEAPER ELECTRICITY DURING CURTAILMENT EVENTS.
- FREE INSTALLATION APPLIANCE INSTALLATION.
- REDUCED FUEL POVERTY.
- NO RISK TARIFFS

PROTECTIONS:

- NO PEAK PENALTIES
- NO EXPOSURE TO PRICE SPIKES
- COMFORT CONSTRAINTS ENFORCED

Adaptive Management

CHALLENGE	THREAT LEVEL	STRATEGY MODIFICATION & MITIGATION
INCREASED CURTAILMENT PRESSURE	INABILITY TO CURTAIL AMOUNT OF EXCESS ENERGY THAT IS BEING PRODUCED, REDUCING SYSTEM EFFECTIVENESS.	LOWERING FEES AND SCHEME EXPANSION: IN THE CASE OF A SUDDEN HIGH PERIOD OF PRESSURE, WE OFFER SIGNIFICANT DISCOUNTS AND COST SAVINGS FOR PARTICIPANTS. IN THE CASE OF A LONG PERIOD OF INCREASED CURTAILMENT PRESSURE, WE EXPAND THE SCHEME TO REGIONS OUTSIDE OF ORKNEY, ALLOWING US TO CURTAIL MORE ENERGY.
PARTICIPATION UNCERTAINTY	NOT ENOUGH HOUSEHOLDS PARTICIPATE IN SCHEME, PREVENTING US FROM MEETING MINIMUM AMOUNT OF CURTAILED ENERGY.	DUAL STRATEGY: INCREASE COST SAVINGS FOR USERS OF OUR SCHEME, HEIGHTENING THE BENEFIT OF PARTICIPATION. ALSO EXPAND THE SMART PRODUCTS THAT WE PROVIDE - FOR INSTANCE SMART REFRIGERATORS AND DISHWASHERS. THIS WOULD ALLOW MORE ENERGY TO BE CURTAILED PER HOUSEHOLD.
DECREASE IN FUNDING	INABILITY TO CARRY OUT THE PROPOSAL IN FULL DUE TO REDUCED BUDGET	ENHANCEMENT OF THE COMMUNITY FUND: TO MITIGATE THE CHALLENGES OF A REDUCED BUDGET WHILE ADDRESSING DISTRIBUTIONAL FAIRNESS, THE PROJECT WILL INCLUDE AN ENHANCEMENT OF THE COMMUNITY FUND THROUGH A 20% COMMUNITY DIVIDEND POOL. 20% OF TOTAL SYSTEM SAVINGS WILL BE DIVERTED AWAY INTO A FUND MANAGED BY THE ORKNEY COMMUNITY TRUST. THE COMMUNITY FUND TRUST ENSURES THAT LOW FLEXIBILITY HOUSEHOLDS, THOSE WITHOUT HIGH-DRAIN ASSETS LIKE EVS OR LARGE HEATERS, STILL CAPTURE THE BENEFITS OF THE PROJECT THROUGH COMMUNITY INITIATIVES. INITIATIVES WOULD CENTER AROUND SPREADING THE REWARDS SUCH AS FUEL MITIGATION SUPPORT.
REGULARITY EMPHASIS ON FAIRNESS	REGULATORY ROADBLOCKS PREVENTING GROWTH, EXPANSION & COLLABORATION WITH STAKEHOLDERS	

AI Usage Disclaimer

Generative AI models, namely Gemini and Claude, were used in the project for code generation for data analysis and plotting of graphs, as well as image generation for slides (as seen on this slide). All of the final analysis and arguments are of our own. AI was only leverage to improve clarity of expression and flow of text in the appendix slides.



AI renders



References

National Records of Scotland. (2024). Council area profile: Orkney Islands. <https://www.nrscotland.gov.uk/statistics-and-data/council-area-profiles/orkney-islands/>

European Marine Energy Centre. (2024, May 22). Orkney household energy survey highlights growing energy costs and interest in renewables. <https://www.emec.org.uk/orkney-household-energy-survey-2024/>

Islands Centre for Net Zero <https://www.icnz.org/island-profiles/orkney>

Eurostat/GOV.UK (2025/26): "Household Energy Efficiency Statistical Release," providing data on the uptake of heating controls and smart meters (approx. 14%–16% for energy-specific IoT). 15% market penetration for energy devices that have financial incentive for initial pilot launch. **max 3000**

Boiler Guide. (2025, July 29). Electric boilers vs. gas boilers: Pros, cons & costs (2025). <https://www.boilerguide.co.uk/compare/types/electric-vs-gas>

Stakeholders hold different priorities

